

HRMS Leave Management

Business Test Case Documentation

Leave Balance • Carry Forward • Rollback

Comprehensive functional, negative, boundary, data-integrity and workflow coverage

Purpose

This document defines business-oriented test cases for the current Leave Balance and Carry Forward V1 implementation. It is designed for QA, HR, managers, developers and business stakeholders.

Test case structure

Each case identifies the rule, setup/example, action, expected result and priority. The suite includes positive, negative, boundary, rerun, transaction, rollback, API, data-isolation and regression scenarios.

Version: Carry Forward V1

1. Business Rules Under Test

Rule ID	Business Rule
BR-01	Available balance = Entitled Days + Carry Forward Days - Used Days.
BR-02	Carry forward is controlled by LeavePolicyRule.IsCarryForwardAllowed.
BR-03	Carry forward cannot exceed MaximumCarryForwardDays.
BR-04	Closing balance = MAX(0, Entitled Days - Used Days).
BR-05	If carry forward is not allowed, CarryForwardDays must be 0.
BR-06	Target-year annual entitlement comes from the applicable policy rule.
BR-07	New target-year entitlement starts with UsedDays = 0.
BR-08	An active employee/year/type entitlement must not be duplicated.
BR-09	Generation is transactional; a failed generation must not leave partial committed generation.
BR-10	Rollback uses soft delete for eligible carry-forward records.
BR-11	Rollback must not remove a carry-forward balance that has already been used.
BR-12	Leave types with zero carry-forward are not affected by carry-forward rollback.
BR-13	After rollback, generation can create a new active entitlement.
BR-14	Balance API exposes EntitledDays, CarryForwardDays, UsedDays and AvailableDays.
BR-15	Deleted entitlement records do not participate in normal balance/generation processing.

2. Baseline Test Data

Employee	Example ID	Purpose
Employee 1 / Anzar	19ED1DF1-C3CC-4614-85F0-08DEF7B598A7	Balance and carry-forward examples
Employee 2 / John	5D09DE86-0892-44E3-FEB0-08DEECB71347	Second employee / partial carry example

Leave Type	Annual	Carry Allowed	Maximum
CL	12	No	0
EL	15	Yes	10
PL	10	Yes	5
SL	12	No	0

Target year used in examples: 8BB8836A-3920-4DA2-41CF-08DEFD305CC8. Replace IDs for another environment.

3. Leave Balance Test Cases

3.1 Core balance calculation

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
BAL-001	Normal balance	12 entitled, 0 carry, 0 used	GET balance	Available = 12	P1
BAL-002	Approved leave reduces balance	12, 0, 3 used	GET balance	Available = 9	P1
BAL-003	Balance includes carry	10 entitled, 5 carry, 0 used	GET balance	Available = 15	P1
BAL-004	Carry plus usage	10, 5, 2 used	GET balance	Available = 13	P1
BAL-005	All leave types	Employee has CL/EL/PL/SL	GET without type filter	All active balances returned	P1
BAL-006	Single type filter	Valid CL ID	GET with leaveTypeId	Only CL returned	P1
BAL-007	Zero balance	12 entitled, 0 carry, 12 used	GET balance	Available = 0	P1
BAL-008	Carry fully consumed	10 entitled, 5 carry, 15 used	GET balance	Available = 0	P1
BAL-009	Carry partially consumed	10, 5, 12 used	GET balance	Available = 3	P1
BAL-010	Deleted record excluded	Active + IsDeleted=1 records	GET balance	Deleted record ignored	P1

3.2 Boundary and invalid balance data

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
BAL-011	Decimal calculation	12.50 entitled, 2.25 carry, 1.25 used	GET balance	Available = 13.50	P2
BAL-012	All zeros	0/0/0	GET balance	Available = 0	P2
BAL-013	Invalid negative entitlement	Test fixture has -1	GET balance	Invalid data is not silently accepted; defect/validation identified	P1

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
BAL-014	Invalid employee	Random GUID	GET balance	No unrelated data returned	P1
BAL-015	Invalid year	Random GUID	GET balance	No unrelated data returned	P1
BAL-016	Invalid type filter	Random type GUID	GET balance	No unrelated type returned	P2
BAL-017	Year isolation	Employee has FY2026 and FY2027	Request FY2027	Only FY2027 returned	P1

4. Carry Forward Calculation Test Cases

4.1 Core rules

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
CF-001	Full PL carry to maximum	PL 10, used 0, max 5	Generate	Carry = 5	P1
CF-002	Partial PL carry	PL 10, used 7, max 5	Generate	Carry = 3	P1
CF-003	CL cannot carry	CL 12, used 0, allowed No	Generate	Carry = 0	P1
CF-004	SL cannot carry	SL 12, used 4, allowed No	Generate	Carry = 0	P1
CF-005	Full EL carry	EL 15, used 0, max 10	Generate	Carry = 10	P1
CF-006	Partial EL carry	EL 15, used 8, max 10	Generate	Carry = 7	P1
CF-007	All balance used	PL 10, used 10	Generate	Carry = 0	P1
CF-008	Used exceeds entitlement fixture	Entitled 10, used 12	Generate	Closing cannot be negative; Carry = 0	P1
CF-009	Closing equals maximum	Closing 5, max 5	Generate	Carry = 5	P2
CF-010	Closing zero	Entitled 10, used 10	Generate	Carry = 0	P2
CF-011	Maximum zero	Allowed Yes, max 0	Generate	Carry = 0	P2
CF-012	Decimal carry	10.50 entitled, 3.25 used, max 5	Generate	Carry = 5.00	P2

4.2 Target-year creation

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
CF-013	Target record created	Valid source + target + rule	Generate	Target entitlement created	P1
CF-014	Target annual entitlement	Target policy annual = 15	Generate	Target EntitledDays = 15	P1
CF-015	Target usage starts zero	Source UsedDays = 7	Generate	Target UsedDays = 0	P1
CF-016	Calculated carry stored	Closing 3, max 5	Generate	Target CarryForwardDays = 3	P1
CF-017	Employee preserved	Source belongs to Employee A	Generate	Target belongs to Employee A	P1
CF-018	Leave type preserved	Source is PL	Generate	Target is PL	P1
CF-019	No source entitlement	Employee has no previous-year record	Generate	No invented carry record	P1

5. Leave Year / Source Selection Tests

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
CFY-001	Correct previous year	Several prior years exist	Generate FY2027	Immediately previous valid year selected	P1
CFY-002	Company isolation	Another company has a closer year	Generate	Other company's year ignored	P1
CFY-003	No previous year	First year for company	Generate	Meaningful error; no records created	P1
CFY-004	Future year exists	FY2028 exists	Generate FY2027	FY2028 never used as source	P1
CFY-005	Target year invalid	Random target GUID	Generate	Not-found/error; no writes	P1
CFY-006	Target year deleted	Target IsDeleted=1	Generate	Deleted target rejected	P1
CFY-007	Previous year deleted	Prior candidate IsDeleted=1	Generate	Deleted source ignored	P1
CFY-008	Late source leave	Leave approved before generation	Generate after close	Latest source UsedDays affects carry	P1

6. Idempotency / Rerun Tests

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
IDM-001	First generation	No active target entitlement	Generate once	One active target record	P1
IDM-002	Second generation	Target already active	Generate again	No duplicate active record	P1
IDM-003	Multiple reruns	Target active	Generate 3+ times	Still one active record	P1
IDM-004	Existing manual target record	Target already active	Generate	Existing record not duplicated/overwritten	P1
IDM-005	Generate after rollback	Old target is soft-deleted	Generate	New active record created	P1
IDM-006	Soft-deleted + active history	Old deleted + new active	Generate	No second active duplicate	P1
IDM-007	Source balance changes before first generation	UsedDays changes	Generate	Latest closing balance used	P1

7. Transaction / Failure Tests

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
TX-001	Successful generation	Valid data	Generate	All intended records committed	P1
TX-002	Validation failure	Invalid year/rule	Generate	No partial target records	P1
TX-003	Mid-run failure	Force one processing failure	Generate	Transaction rolls back per current design	P1
TX-004	Database error	Simulate DB failure	Generate	No partial committed generation	P1
TX-005	Cancellation	Cancel request during processing	Generate	No unintended partial committed state	P2

8. Policy Configuration Tests

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
POL-001	Carry flag false	Allowed = 0	Generate	Carry = 0	P1
POL-002	Carry flag true	Allowed = 1, max 5	Generate	Carry calculated/capped	P1
POL-003	Maximum changed	Max 5→7 before generation	Generate	New maximum used	P1
POL-004	Maximum below closing	Closing 8, max 5	Generate	Carry = 5	P1
POL-005	Maximum above closing	Closing 3, max 5	Generate	Carry = 3	P1
POL-006	Rule/type mismatch	PL rule with CL type	Create/generate	Operation rejected	P1
POL-007	Missing rule	Referenced rule absent	Generate	Not-found error	P1
POL-008	Deleted rule	Rule IsDeleted=1	Generate	Rule not used	P1
POL-009	Target annual entitlement changed	Target annual = 18	Generate	Target annual = 18	P1

9. Rollback Tests

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
RB-001	Rollback unused carry	Carry 5, Used 0, active	Rollback	IsDeleted = 1	P1
RB-002	Rollback multiple employees	Several eligible records	Rollback	All eligible records soft-deleted	P1
RB-003	CL unaffected	CL Carry = 0	Rollback	CL remains active	P1
RB-004	SL unaffected	SL Carry = 0	Rollback	SL remains active	P1

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
RB-005	Rollback after usage	Carry 5, Used 2	Rollback	Rejected; record remains active	P1
RB-006	One used candidate	Several candidates, one Used > 0	Rollback	No unintended eligible deletions when operation rejects	P1
RB-007	Invalid target year	Random GUID	Rollback	Error; no changes	P1
RB-008	Already rolled back	No active carry records	Rollback	Safe/no-op; no corruption	P2
RB-009	History preserved	Previously active carry record	Rollback	Row remains with IsDeleted=1	P1
RB-010	Regenerate after rollback	Rollback completed	Generate	New active record created	P1
RB-011	UsedDays preserved	Eligible Used=0	Rollback	UsedDays unchanged	P1
RB-012	EntitledDays preserved	Eligible record	Rollback	Historical EntitledDays unchanged	P2

10. API Contract Tests

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
API-001	Valid generate request	Valid LeaveYearId	POST generate	2xx + success message	P1
API-002	Empty generate body	{}	POST generate	Validation/error response	P1
API-003	Malformed GUID	Invalid text GUID	POST generate	400/validation response	P1
API-004	Valid rollback	Valid LeaveYearId	POST rollback	2xx + success message	P1
API-005	Empty rollback body	{}	POST rollback	Validation/error response	P1
API-006	Valid balance request	Valid employee/year	GET balance	2xx + correct DTO	P1
API-007	Invalid employee	Random GUID	GET balance	No unrelated data	P1
API-008	Invalid year	Random GUID	GET balance	No unrelated data	P1
API-009	Leave type filter	Valid type ID	GET balance	Only requested type	P1
API-010	Wrong HTTP verb	GET on POST route	Call endpoint	Routing/method error	P2

11. Employee Isolation Tests

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
EMP-001	Employee A receives only A's carry	A and B have different balances	Generate	Each gets own calculation	P1
EMP-002	B generation does not alter A	Separate employees	Generate B	A unchanged	P1
EMP-003	Same leave type across employees	A PL + B PL	Generate	Separate entitlement rows	P1
EMP-004	Multiple types per employee	A has CL/EL/PL/SL	Generate	Each policy applied independently	P1
EMP-005	Employee without source	No prior-year entitlement	Generate	No invented carry	P1

12. Database Integrity Tests

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
DB-001	Duplicate active entitlement	Same employee/year/type twice	Create	Business/unique rule prevents duplicate active row	P1
DB-002	Soft-deleted history	Old row IsDeleted=1	Generate	New active row allowed	P1
DB-003	Deleted row excluded from balance	IsDeleted=1	GET balance	Ignored	P1
DB-004	Deleted source excluded	Previous row IsDeleted=1	Generate	Not used as source	P1
DB-005	Decimal precision	Decimal annual/max	Generate	Values preserved to 2 decimals	P2
DB-006	Rollback audit	Eligible row	Rollback	Deletion/audit fields updated	P1

13. Regression Tests Around Leave Operations

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
REG-001	Apply leave after generation	Valid balance	Apply	Request created without corrupting entitlement	P1
REG-002	Approve leave	Pending 2 days	Approve	UsedDays increases by 2	P1
REG-003	Reject leave	Pending leave	Reject	UsedDays unchanged	P1
REG-004	Cancel approved leave	Approved leave	Cancel	UsedDays restored per existing rules	P1
REG-005	Balance after approval	Carry exists	Approve 2	Available decreases by 2	P1
REG-006	Rollback after use	Carry record Used > 0	Rollback	Rejected	P1
REG-007	Generate after source usage changes	Source usage updated	Generate	Latest closing used	P1
REG-008	Generation must not reset existing target usage	Active target has usage	Generate	Existing target not overwritten	P1

14. End-to-End Scenarios

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
E2E-001	Maximum PL carry	10 entitled, 0 used, max 5	Close → Generate → Balance	15 available	P1
E2E-002	Partial PL carry	10 entitled, 7 used, max 5	Close → Generate → Balance	13 available	P1

TC ID	Scenario / Rule	Preconditions / Example	Action	Expected Result	Priority
E2E-003	No CL carry	12 entitled, 0 used, carry disabled	Close → Generate → Balance	12 available	P1
E2E-004	Use carried leave	Target PL has 5 carry	Approve 3 days	Used 3; available 12	P1
E2E-005	Operational correction	Generated but unused	Rollback → Generate	History retained; new active row	P1
E2E-006	Rollback blocked after use	Generated then leave approved	Rollback	Rejected; balance preserved	P1

15. Negative Test Catalog

- Invalid/missing employee, leave year or leave type identifiers.
- Target year does not exist or is deleted.
- Previous year does not exist.
- Previous year belongs to another company.
- Policy rule is missing, deleted or belongs to another leave type.
- EntitledDays is negative.
- Duplicate active entitlement exists.
- Rollback requested after carry-forward has been consumed.
- Database failure occurs during multi-employee generation.
- Concurrent generation and rollback occur.
- Deleted records accidentally participate in balance calculation.
- One employee accidentally receives another employee's entitlement.

16. Boundary Value Analysis

Value	Boundary	Example	Expected
EntitledDays	0	0	No negative value; balance arithmetic correct
UsedDays	0 / equals entitlement	0 or 10 of 10	Closing = 10 or 0
Maximum carry	0 / equals closing	0 or 5	Carry capped correctly
Closing balance	0 / large positive	0 or 100	Carry never exceeds policy max
Employee count	1 / many	1 or all employees	No cross-employee mixing
Leave types	1 / all	PL or CL+EL+PL+SL	Each rule applied independently

17. Priority Definitions

Priority	Meaning
P1 / Critical	Must pass before release. Failure affects entitlement, balance, data integrity or workflow.
P2 / High	Important functional, boundary or operational behavior.
P3 / Medium	Secondary validation or non-critical behavior.

18. QA Execution Checklist

- Confirm test database and target environment.
- Record source-year entitlement values before generation.
- Record policy rule values before generation.
- Execute generation and capture API response.
- Query the database and compare expected versus actual CarryForwardDays.
- Call Balance API and verify AvailableDays.
- Repeat generation and verify no duplicate active records.
- For rollback tests, verify UsedDays before rollback.
- Execute rollback and verify IsDeleted and audit fields.
- Regenerate after rollback and verify a new active record.
- Run approval/cancellation/rejection regression tests.
- Restore controlled test data after destructive test scenarios.

19. Recommended Execution Order

Phase 1 → Policy/master data
Phase 2 → Employee entitlements
Phase 3 → Balance API
Phase 4 → Carry-forward calculations
Phase 5 → Idempotency / rerun
Phase 6 → Transaction failure
Phase 7 → Rollback
Phase 8 → Generate after rollback
Phase 9 → Leave approval/rejection/cancellation regression
Phase 10 → End-to-end business sign-off

20. Business Sign-off Criteria

- All P1 test cases pass.
- No duplicate active entitlement can be produced.
- No employee receives another employee's balance.
- Carry-forward never exceeds the policy maximum.
- No-carry leave types always carry zero.
- Available balance is correct for tested combinations.
- A failed generation leaves no partial committed generation.
- Rollback cannot remove an already-used carry-forward balance.
- Rollback preserves history through soft delete.
- Generate → Rollback → Generate Again works correctly.
- Existing leave approval/cancellation behavior remains correct.

21. API Test Payloads

Generate

POST /api/EmployeeLeaveEntitlement/carry-forward

```
{
  "leaveYearId": "8bb8836a-3920-4da2-41cf-08defd305cc8"
}
```

Rollback

POST /api/EmployeeLeaveEntitlement/carry-forward/rollback

```
{
  "leaveYearId": "8bb8836a-3920-4da2-41cf-08defd305cc8"
}
```


Balance

GET /api/EmployeeLeaveEntitlement/
19ed1df1-c3cc-4614-85f0-08def7b598a7/
8bb8836a-3920-4da2-41cf-08defd305cc8/balance

22. Test Execution Template

TC ID	Actual Result	Status	Defect ID	Executed By	Date	Remarks
CF-001		Not Run				
CF-002		Not Run				
CF-003		Not Run				
RB-001		Not Run				
RB-005		Not Run				
IDM-002		Not Run				
TX-003		Not Run				
E2E-001		Not Run				

23. Final Coverage Summary

Area	Coverage
Balance	Normal, zero, decimal, carry, usage, filters and deleted records
Carry rules	Allowed/not allowed, maximum, full, partial, zero and decimal
Year selection	Previous year, same company, deleted/future/invalid year
Idempotency	Repeated generation, existing target, post-rollback generation
Transactions	Success, validation failure, DB failure and cancellation
Rollback	Eligible, used balance, multiple employees, repeat and soft delete
API	Valid/invalid payloads, identifiers and endpoint contract
Isolation	Employee and leave-type separation
Database	Unique active records, soft delete, audit and precision
Regression	Approval, rejection, cancellation and balance
E2E	Complete business lifecycle

Note: This specification reflects the current V1 business behavior discussed and tested in the project. Exact authentication/authorization roles and future CarryForwardRun audit behavior are not defined here and should receive separate tests when those features are introduced.